

Molecular Analyst / Fingerprinting Plus Quick Guide



Molecular Analyst[®] Software

Fingerprinting Plus
Version 1.12
© Applied Maths 1992-1994

BIO-RAD

**Bio-Rad
Laboratories**

Bio-Rad demo version - NOT FOR SALE

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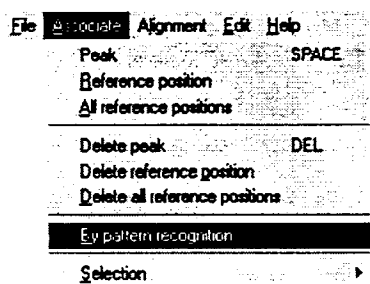
1. What you should know before starting

This quick guide is not designed to replace the user manual for Molecular Analyst Fingerprinting Plus. Many useful and interesting features of the software are only described in the user manual. This guide is only intended to be used as a "Getting started" reference.

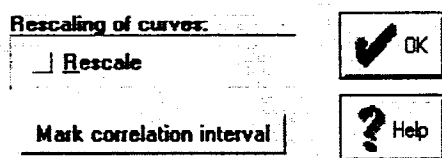
In the sections that follow, all menu commands and button text are shown in *bold italics*. Key terms and dialog box names are shown in *italics*. Submenus are separated from parent menus by vertical slashes (/). Button text is always given between < and > characters.

Examples:

The following menu selection is indicated in the text as *Associate/By pattern recognition*.



The following buttons are indicated as *Rescale* (check box), *<Mark correlation interval>*, *<OK>*, *<Help>*.



2. Introduction to Fingerprinting Plus

The Fingerprinting Plus software is composed of three functional units that work as independent programs and interact with each other through a structured directory and file system. They constitute the three major steps involved in processing a gel or set of profiles, and are accessed through the **Molecular Analyst Fingerprinting Startup program**. They are:

- 1) **Convert**. The Convert function allows you to read the native scanning file, convert it to Fingerprinting Plus format, make the profile length ("resolution") uniform, and add specific lane information. The result is a Fingerprinting Plus file of unnormalized ("raw") tracks documented with lane information fields.
- 2) **Normalize**. The Normalize function is a very important step in the process of creating a Fingerprinting Plus database. Normalization means correcting shifts of whatever kind or origin within and between gels, so that bands or peaks of the same size and type have the same

physical position, even if they occur on different gels. The result is a normalized and documented Fingerprinting Plus database file, ready for analysis and comparison with other database files.

- 3) **Analyze.** This is the final and most interesting part of the software. Analysis involves calculating dendrograms to reveal groups of related patterns, identifying new and unknown patterns against the database or against custom libraries, quantifying band sizes and concentrations, performing principal component analysis, and other functions.

This multistep approach has some key advantages:

- 1) Every step is saved in a separate file. This leaves a trail for control purposes and backup. Whatever happens to a processed gel file, you can always go back to the saved file from the previous step without having to restart from the beginning.
- 2) It gives maximum flexibility with respect to importing and converting different file types.

It is important to note that the multistep flow is unidirectional. For example, if you normalize a gel and change the lane information later in the Analyze program, these changes will not automatically reappear when you want to renormalize the gel, since the raw gel file is different from the normalized gel file.

This Quick Guide describes the use of Fingerprinting Plus in five main sections:

- Installing the software
- Starting and setting up the software
- Converting gels
- Normalizing gels
- Basic Analysis functions

3. Installing the software

- 1) Turn off the computer. If a printer is connected to the LPT1 port of the computer, turn off the printer and disconnect the printer cable from the computer. Insert the protection key (dongle) in the LPT1 port of the computer. Reconnect the printer to the key (if necessary), and switch on the power of the printer and the computer.
- 2) Insert the Molecular Analyst Fingerprinting Plus installation disk 1 into the disk drive.
- 3) In Windows, press "Start" in the Taskbar and select "Run..."
- 4) Type "A:INSTALL" if the disk drive is A: and press <ENTER>.
- 5) An input box shows the directory where Molecular Analyst Fingerprinting Plus will be installed. The default path name may be changed if desired.
- 6) Press the <Start> button to begin the installation.

The installation program creates a program group called "Molecular Analyst 1.12," which contains three icons.

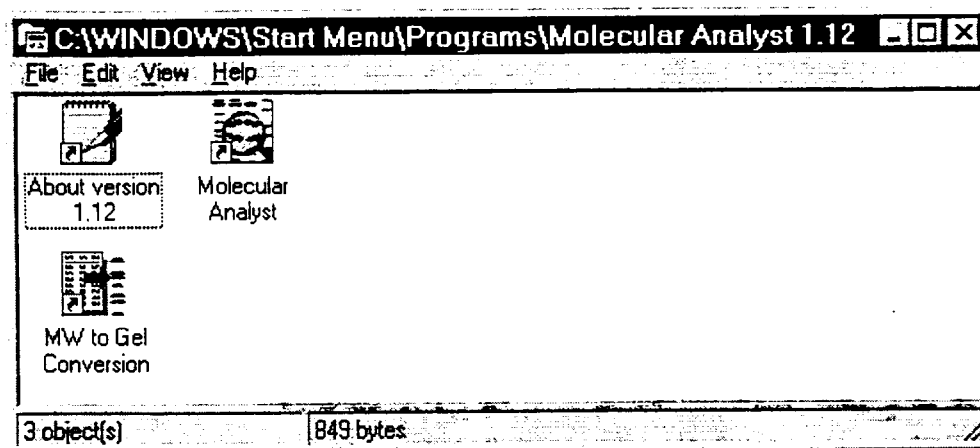


Figure 1. Fingerprinting Plus program group.

The "Molecular Analyst" icon loads the Startup program, from which all Fingerprinting Plus applications can be run. The "About version 1.12" icon contains important information about the software in Microsoft Write™ format. If you purchased the Network version, please refer to the reference manual for detailed documentation.

4. Starting and setting up the software

4.1 Fingerprinting Plus Startup program

Double-click on the "Molecular Analyst" icon in the Molecular Analyst 1.12 program group to run the Startup program. This program shows the introduction screen with directory information (1 on the figure). It allows you to run the Fingerprinting Plus applications **<Convert>** (2), **<Normalize>** (3), and **<Analyze>** (4). It also allows you to create users and customize various settings (colors, directories, labels, etc.) for each **<User>** (5), as well as select among users (8).

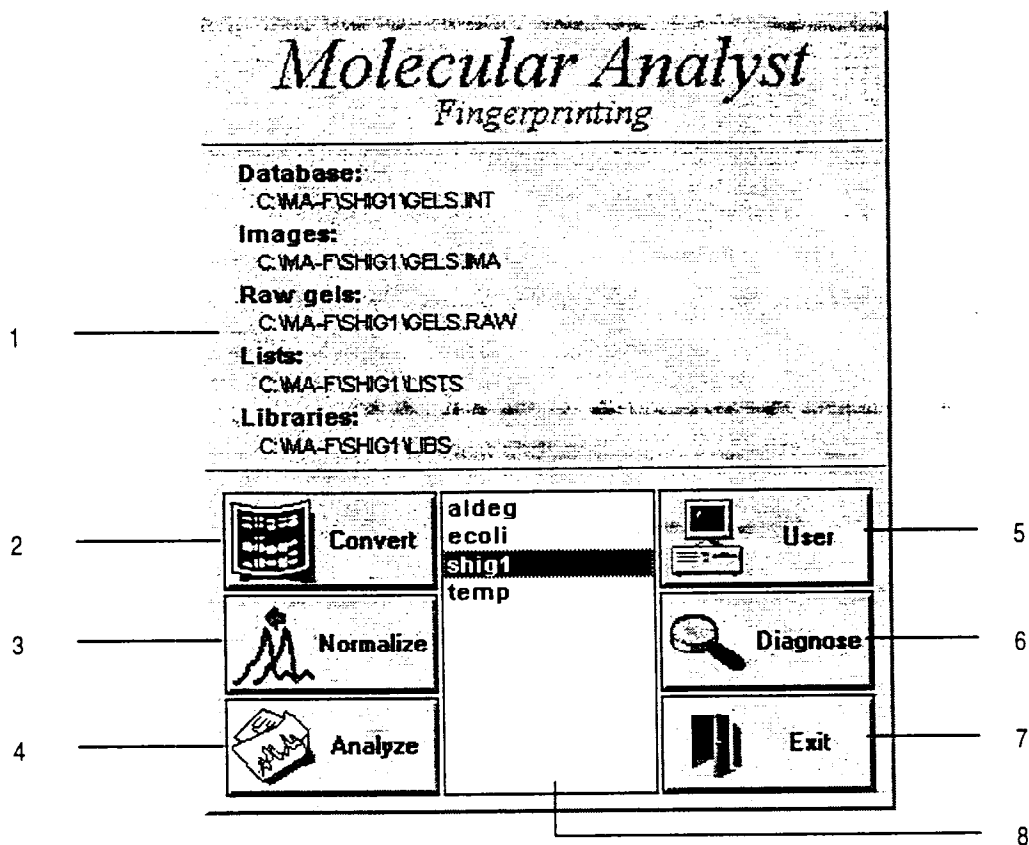


Figure 2. The Fingerprinting Plus Startup program.

<Diagnose> (6) analyzes all the system and hardware features of your computer and detects and reports any errors. It also checks the Molecular Analyst drives and directories of the selected user and inspects the validity of all the files contained therein.

The **<Exit>** button (7) ends your Fingerprinting Plus session.

4.2 Multi-user system

The Fingerprinting Plus system may be used by more than one person and/or for more than one application. This is accomplished by creating a list of Users, each with its own settings, directories, databases, and libraries.

4.2.1 Create a new user by clicking the **<User>** button and *Add new User*. To automatically assign unique directories to the new user, click on **<Yes>** when asked "Automatically create user directories?"

4.2.2 You can further set up and customize the user with *Setup "Username"*.

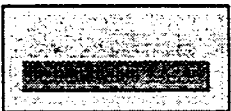
Window color  ☐ Use System Colors R: 67 G: 75 B: 75		Gel colors Background R: 100 G: 100 B: 100 Stain R: 0 G: 0 B: 0		Information field labels Genus Species Serotype Strain no source Comment 1 Comment 2 Names Entry codes Comment lines													
DIRECTORIES <table border="1"> <tr><td>DATABASE</td><td>C:\MA-FIECOLINGELS.INT</td></tr> <tr><td>IMAGES</td><td>C:\MA-FIECOLINGELS.IMA</td></tr> <tr><td>RAW GELS</td><td>C:\MA-FIECOLINGELS.RAW</td></tr> <tr><td>LISTS</td><td>C:\MA-FIECOLINLISTS</td></tr> <tr><td>LIBRARIES</td><td>C:\MA-FIECOLINLIBS</td></tr> <tr><td>COMBINED GELS</td><td>C:\MA-FIECOLINGELS.CMB</td></tr> </table>						DATABASE	C:\MA-FIECOLINGELS.INT	IMAGES	C:\MA-FIECOLINGELS.IMA	RAW GELS	C:\MA-FIECOLINGELS.RAW	LISTS	C:\MA-FIECOLINLISTS	LIBRARIES	C:\MA-FIECOLINLIBS	COMBINED GELS	C:\MA-FIECOLINGELS.CMB
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User "ecoli" Setup ☐ Doublegel normalization ? Help ✓ OK ✗ Cancel																	

Figure 3. User setup window.

- 4.2.3 With **Window color**, you can adjust the RGB (red, green, blue) background color components of all Fingerprinting Plus windows. With **Gel colors**, you can adjust the gel stain and background color components.
- 4.2.4 The **Information field labels** allow you to enter your own names for the seven possible information fields for each pattern in the database. The first three fields are name fields and the fourth and fifth fields are entry code fields. Each pattern may be assigned a unique code, which should be entered in field 4. The last two fields are comment fields, which do not appear in all outputs.
- 4.2.5 The **Directories** are assigned automatically when you select **<Yes>** in the "Automatically create user directories?" box when defining the user. **You should always save TIFF (or other scanning files) for that user in the IMAGES directory.**
- 4.2.6 Doublegel normalization is only for special purposes; see the reference manual.
- 4.2.7 Click on **<OK>** to save the changes made to the user.

5. Converting gels

This section discusses TIFF file importation. Uncompressed TIFF files of 8 bits and 16 bits (with 12-bit OD depth) in PC format are supported by Fingerprinting Plus. To convert other scanning file types, please refer to the reference manual or to the documentation provided with a specific conversion program. Before you can convert a gel, you need to save (or move) the TIFF file to the IMAGES directory of the current user.

- 5.1.1 In the Molecular Analyst Fingerprinting Startup program, click the **<Convert>** button to load the Conversion program.
- 5.1.2 Select **File\Load image** and select the TIFF file. Choose **Normal** image size if the image is not too large, and check **Negative** if the bands are light on a dark background (e.g., ethidium bromide-stained UV captures). Click **<OK>** to load the file.
- 5.1.3 If the gel appears as light bands on a dark background, select **Edit\Negativize**.

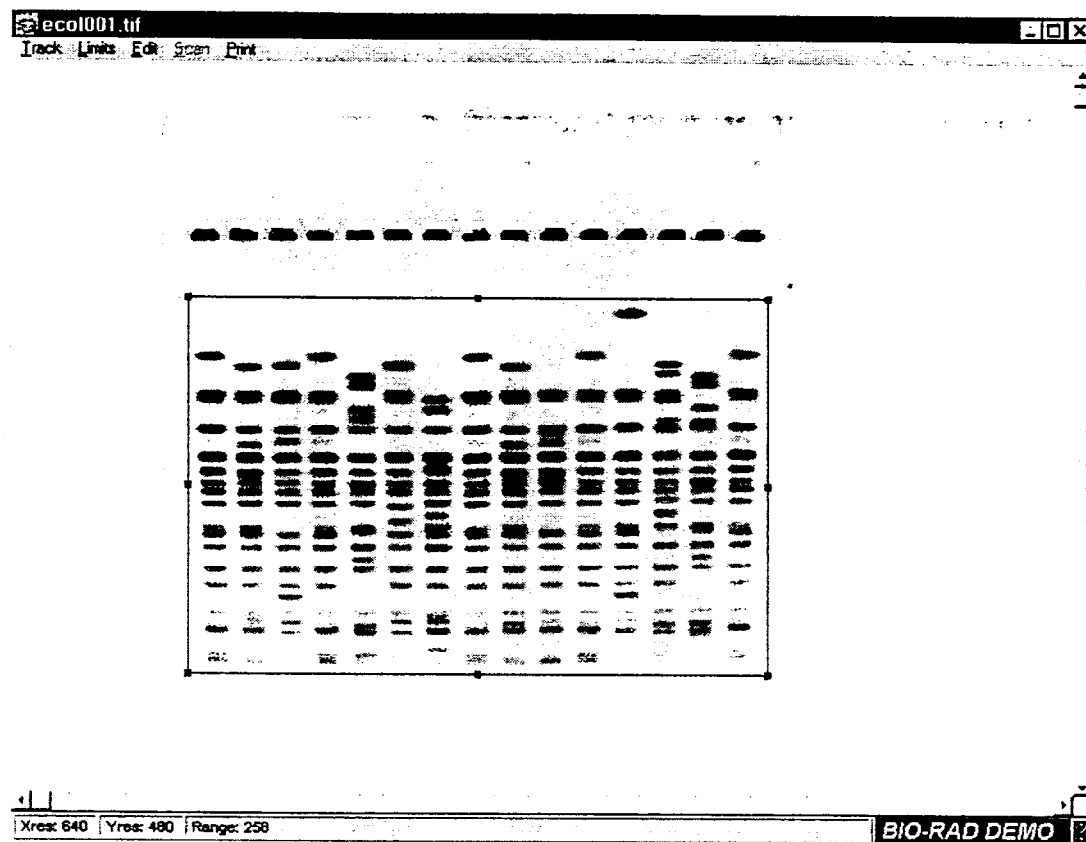


Figure 4. Image conversion window.

The green rectangle delimits the region of interest of the gel; tracks and gelstrips will be extracted within the borders of this rectangle. Adjust the rectangle by dragging the green squares as necessary, and use the SHIFT key to distort the rectangle in each square. The shape of the cursor indicates which directions you can drag.

- 5.1.4 If necessary, select **Edit\Optimize image** to optimize the brightness and contrast within the adjusted bordering rectangle.
- 5.1.5 Select **Edit\Settings** and specify the approximate **Number of tracks** in the **Settings** dialog box. Read the vertical resolution of the gel (**Yres**) in the bottom of the image window and adjust the **Track resolution** to the closest round number smaller than **Yres** (e.g., if **Yres** is 1024, choose 1000). Check **Rescaling\Whole gel** to optimize the curves within the OD range. Set **Curve smoothing** to 0 and **Number of nodes** to 3. Click **<OK>** to implement the changes.

- 5.1.6 Back in the image window, select **Track\Auto search** to find the patterns automatically. Each found lane on the image is represented by a *spline*.
- 5.1.7 Adjust the position of each spline as necessary by grabbing the nodes using the mouse. Use the SHIFT key to distort a spline in one node.
- 5.1.8 Add splines (**Track\Add**) or remove splines (**Track\Remove**) if necessary.
- Each spline has two (inner) black lines and two (outer) blue lines. The black lines show the thickness of the averaging zone for extraction of densitometric curves, while the blue lines show the thickness of the gelstrips that will be extracted from the image.
- 5.1.9 Select **Edit\Settings** again and adjust the **Spline thickness** so that the black lines stay within the center of the bands (covering 1/2 of the band width).
- 5.1.10 Adjust the **Gelstrip thickness** so that the blue lines enclose the complete patterns (blue lines of neighboring patterns should nearly touch each other). See Figure 5 for an optimally adjusted example.

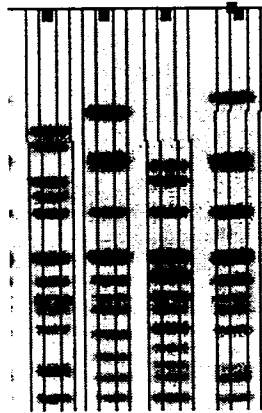


Figure 5. Optimal spline and gelstrip thickness settings, detail.

- 5.1.11 Select **Scan**. When scanning is finished, the tracks are numbered on the image. Move the image window so that the editor window in the background becomes visible. *Do not close the image window, as gelstrip information will be lost!*

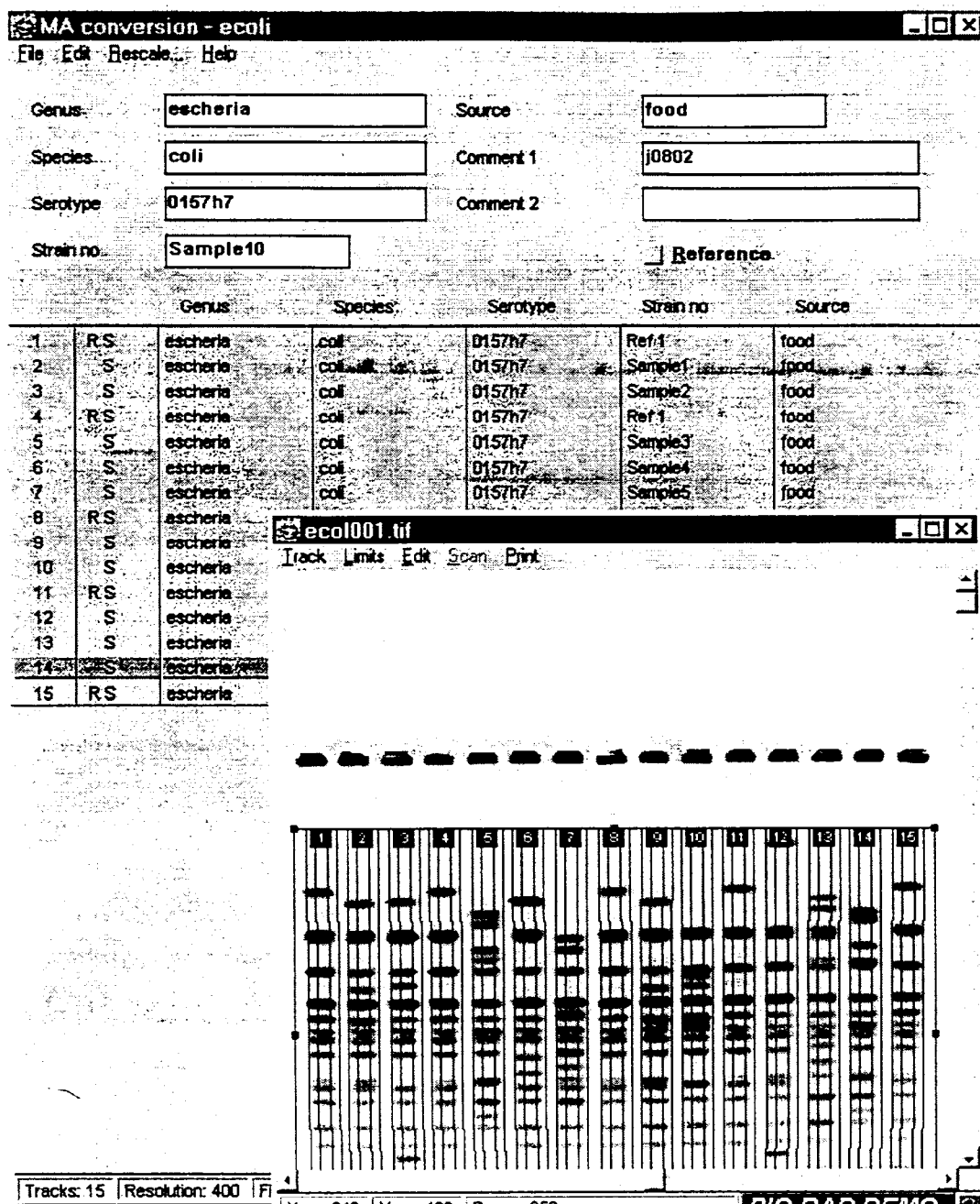


Figure 6. Entering lane information in the Conversion program.

5.1.12 Enter the information fields for each lane, and mark the *reference lanes* (i.e., the lanes that will serve for normalization) by checking the **Reference** checkbox. After entering the information for each lane, accept and validate with RETURN. The information is now shown in the list of lanes.

5.1.13 Finally, save the gel with **File|Save gel**, and exit with **File|Exit**.

6. Normalizing gels

6.1 Setting up the Standard

- 6.1.1 Click the **<Normalize>** button in the Startup program. (The first time you do this, the program will give you the warning message "*Normalization settings not found*" because the settings are not defined yet. Press **<OK>** to continue.)
- 6.1.2 First adjust your settings with **Edit|Settings**. Adjust **Resolution** to be the same as or lower than the settings in the Conversion program. As a general rule, 500 is enough for normal gel types and 1000 is sufficient for long runs (e.g., AFLP). Specify a **Smoothing** range of 5–7, and set **Background subtraction** to 16 using the **Rolling disk** method (be sure to check **Apply**). Select a higher value for patterns with sharp bands or a lower value for blurred patterns. Activate both **Show 2D-gelstrips** and **Save 2D-gelstrips**. Set **Space between tracks** to 3 and disable **Rescale**. Press **<OK>** to save the settings. The program will warn you that "*Changing Normalization settings will cause new gels to be incompatible with previous*"; press **<OK>** to apply.

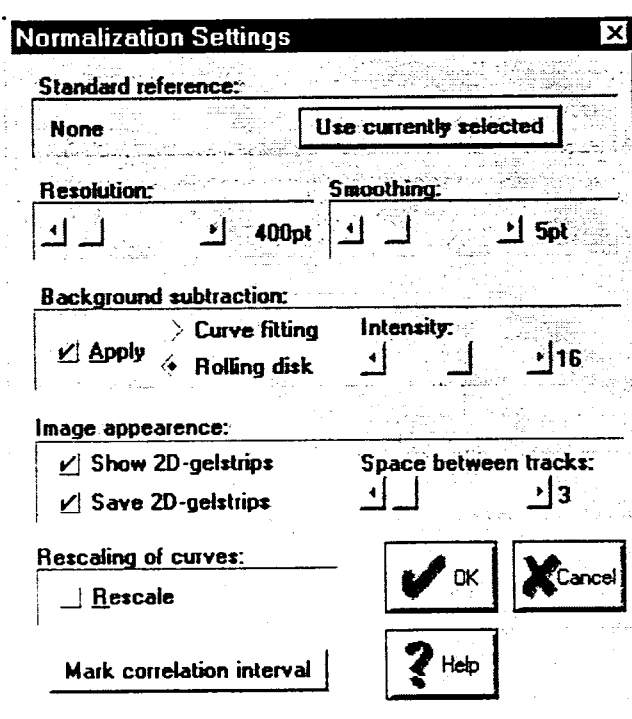


Figure 7. Normalization settings.

- 6.1.3 Load the converted gel with **File|Load**. The image for each lane is displayed as a separate *gelstrip*.
- 6.1.4 Identify the best-looking reference lane (marked with "R") on the gel and click this one with the mouse.

6.1.5 Call *EditSettings* again and click *<Use currently selected>* under *Standard reference*. This lane will serve as the standard for aligning every reference lane on every gel.

6.1.6 Press *<OK>* to save and *<OK>* again to acknowledge the warning message. The *standard* pattern is now displayed on the left-hand side of the window, indicated with *STD*.

Now you are ready to define the bands on the *standard* that will be used for aligning the corresponding bands on the reference patterns. We call these bands the *reference positions*.

6.1.7 Select *EditReference positions* to enter the reference positions editing mode. The status bar shows *DEFINING NEW REFERENCE PEAKS*.

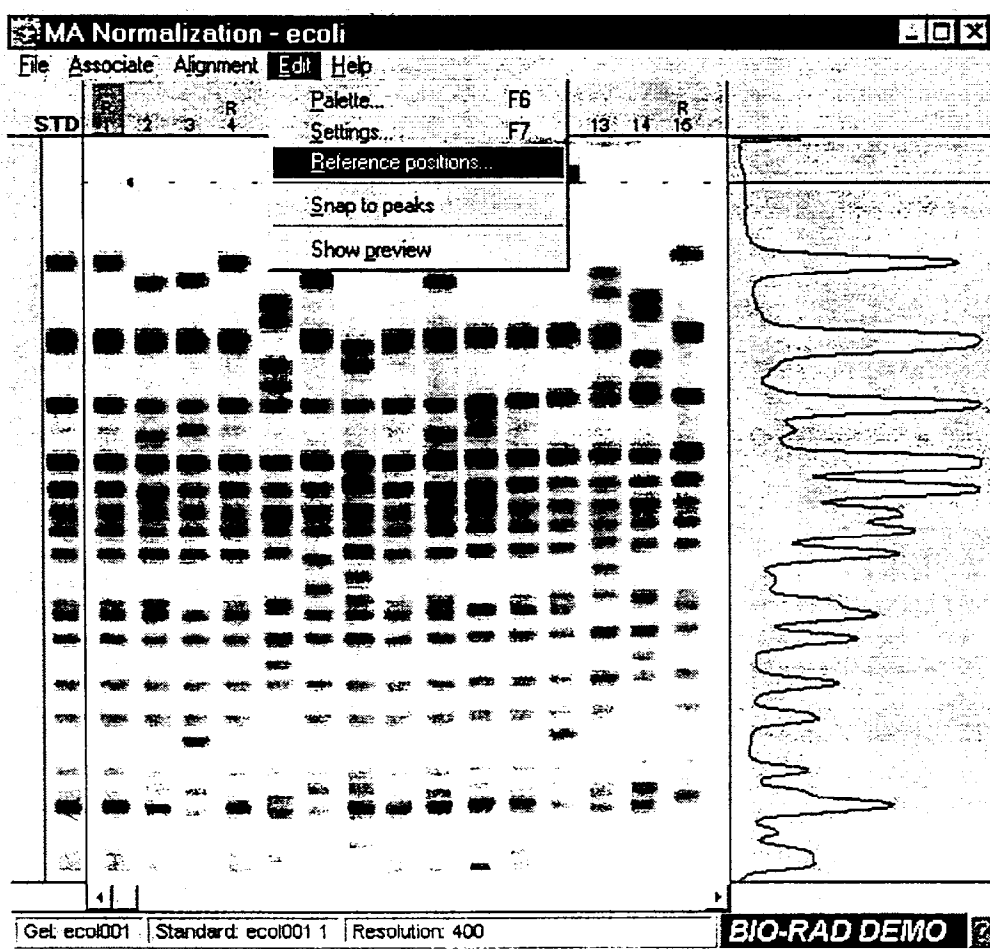


Figure 8. Entering the Reference positions editor.

6.1.8 Click on a clear band on the standard and select *PeakAdd*. A first reference position is defined.

6.1.9 Continue selecting clear bands and *PeakAdd* until all unambiguous bands on the standard are marked as reference positions, or at least until no large intervals are present between the reference positions.

6.1.10 Press <OK> to save the set of reference positions. You are now back in the Normalization window. Do not change the reference positions once you have normalized gels to the database!

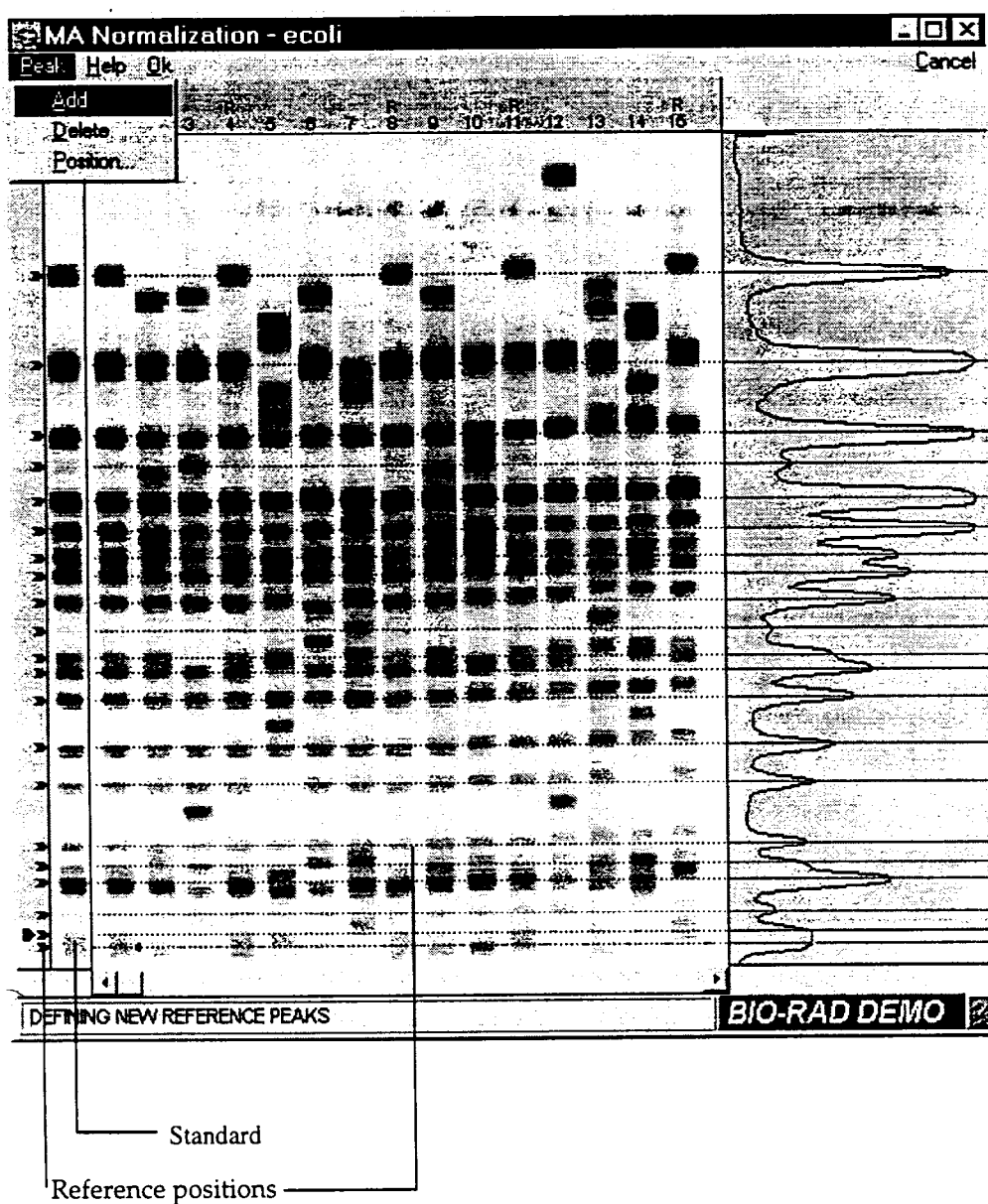


Figure 9. Defining reference positions on the standard pattern.

6.2 Normalizing a gel

A gel is normalized in two steps:

- 1) *Associating* the bands on each reference pattern with the corresponding reference positions on the standard left (see Figure 9), and

- 2) *Aligning* the gel so that all bands on the reference patterns are aligned with their corresponding reference position.
- 6.2.1 Select *Associate* | *By pattern recognition* to automatically find the corresponding bands for each reference position. As a result, all bands along the references will be associated, as shown in Figure 10.

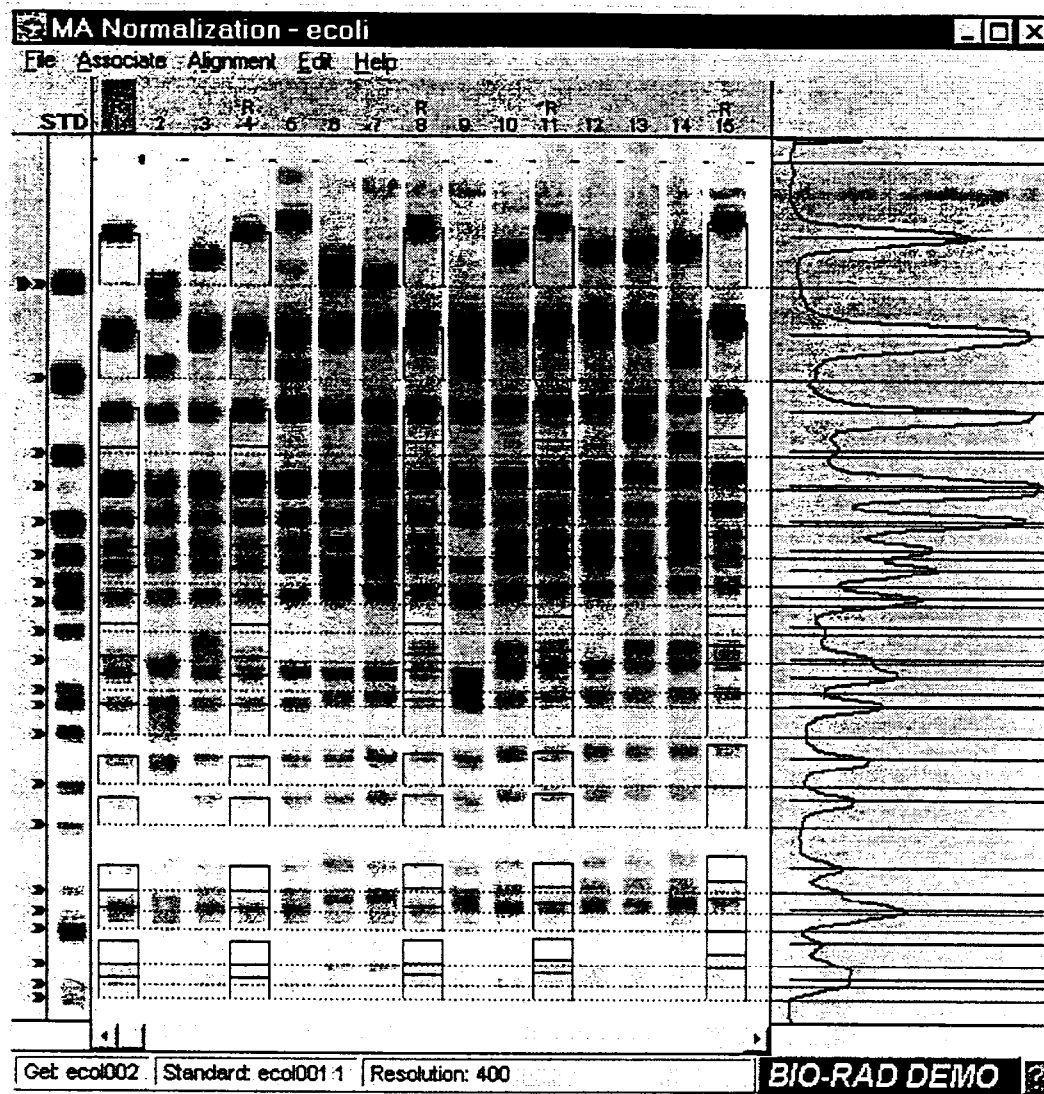


Figure 10. Bands on reference patterns associated with corresponding reference positions.

- 6.2.2 Carefully inspect the gel for incorrectly associated bands.
- 6.2.3 If you find a wrong association, first click the left mouse button on the red dashed line to select the reference position, and then click the right mouse button on the band you want to associate with it.
- 6.2.4 If you feel an associated band is unclear or doubtful, click the left mouse button on it and press DEL or *Associate* | *Delete peak*. A whole reference pattern can be skipped by holding the DEL key until all its associations are removed.

6.2.5 Finally, align the gel with *AlignmentAlign associated peaks*. Select *FileSave* to save the normalized gel to the Fingerprinting Plus database. While saving, the process of subtracting the background from each track is shown.

6.2.6 Exit the program with *FileExit*.

7. Basic Analysis functions

7.1 Main window

7.1.1 In the Molecular Analyst Fingerprinting Startup program, click the *<Analyze>* button.

The *Main window* consists of three panels: the *gel files* (upper left of Figure 11), the *list* of selected patterns (right), and the list boxes showing available list and clusterbases (bottom left). If you have the Database Sharing tools, it contains a fourth panel called the *shared database* (bottom right).

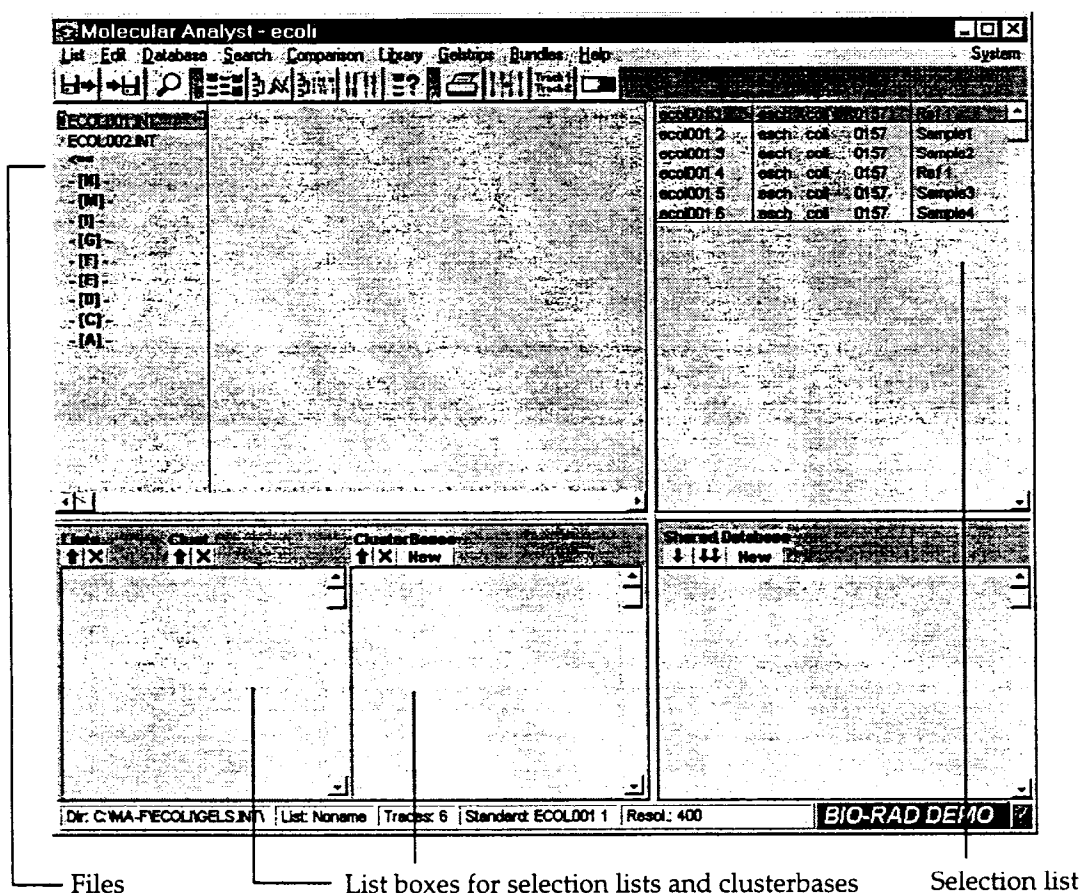


Figure 11. Analyze main window.

7.2 Gel window

- 7.2.1 Double-click on a gel file (or highlight the file name and press RETURN) to open the *gel window*. It shows the information for each pattern. Depending on the window size, more or less fields may be shown or information may be abbreviated.
- 7.2.2 Double-click on a pattern (or *Track/Information*) to show its “identity card”—a gelstrip and all the information fields, which you can edit at this point (Figure 12).
- 7.2.3 Press <OK> to accept changed information on the track.
- 7.2.4 Back in the gel window, select *File/Save* to save the changes.

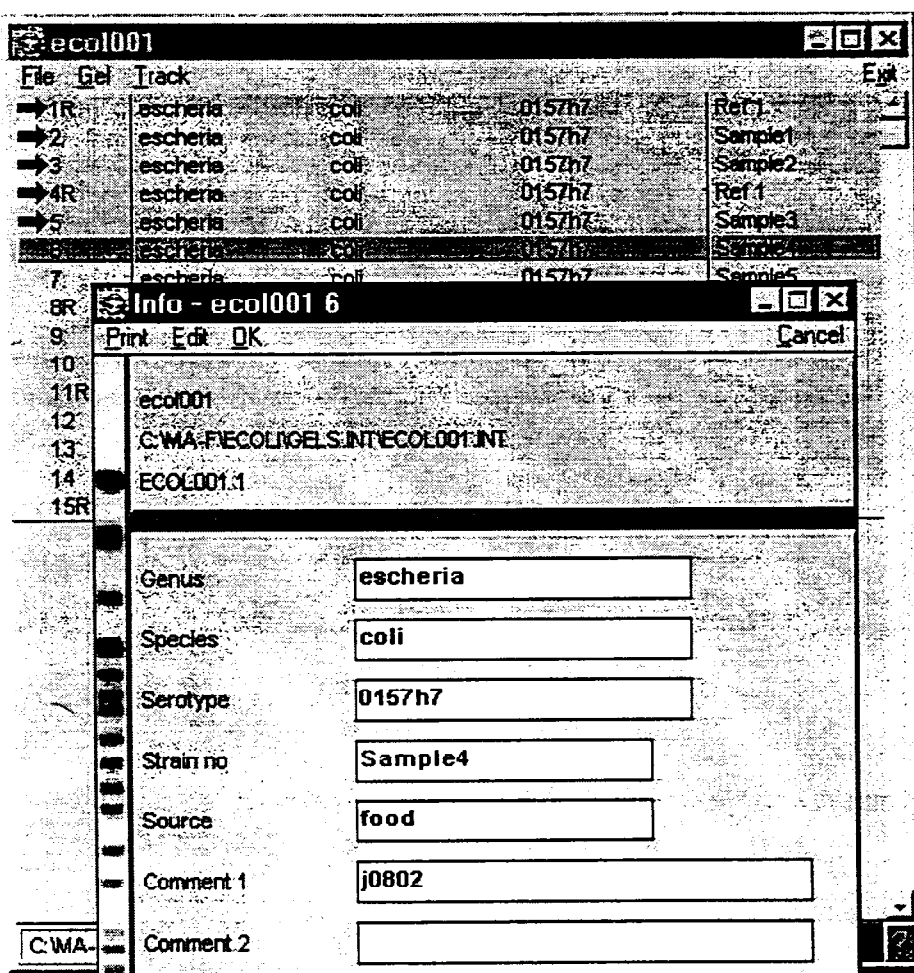


Figure 12. Gel window (back) and identity card with gelstrip of lane 6 (front).

- 7.2.5 Select *Gell2D-gel image* to display an image of the normalized gel.
- 7.2.6 Within the image window, you may select *Print* to open the *Printer manager*. We will come back to the printer manager later.

7.2.7 Select **<Exit>** to close the gel image.

7.3 Defining bands

We will now identify bands on this gel.

7.3.1 In the gel window, select **GelBands** to open the *Band search window* (Figure 13).

7.3.2 Select **Bands\Auto search** to automatically find bands on the gel pattern.

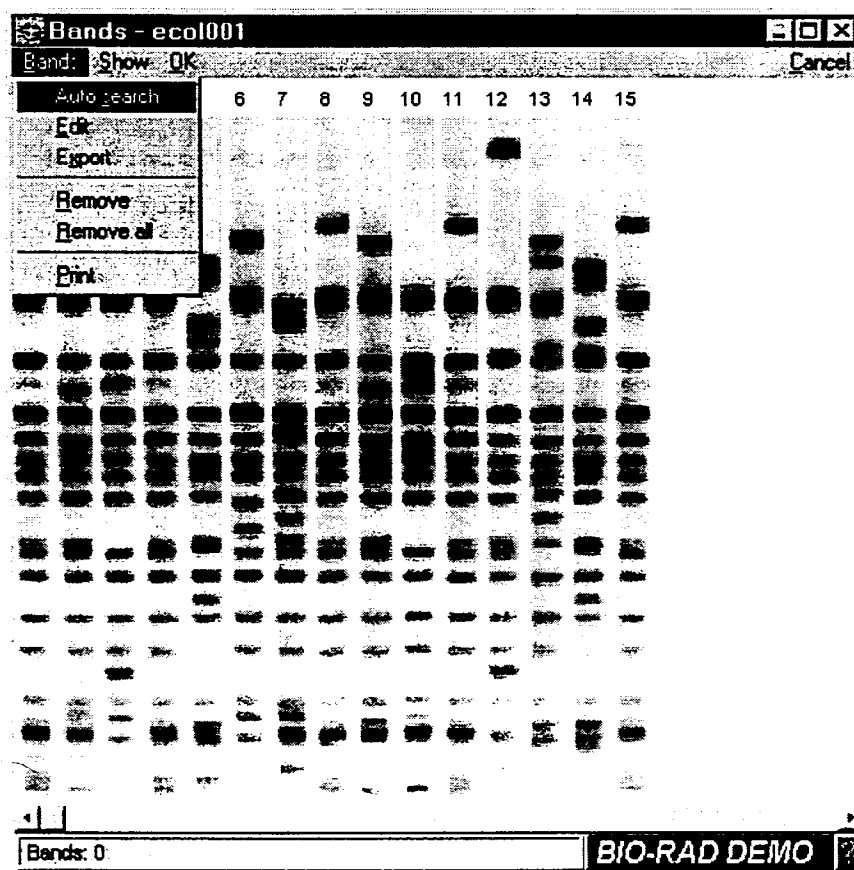


Figure 13. Band search window.

The sensitivity of this search depends on the *band search settings*: If too many false peaks are found or if real bands are undetected, you can change the search sensitivity in the *Band settings* dialog box (Figure 14).

7.3.3 From within any window press the F7 key, or select **System\Band settings** in the main window.

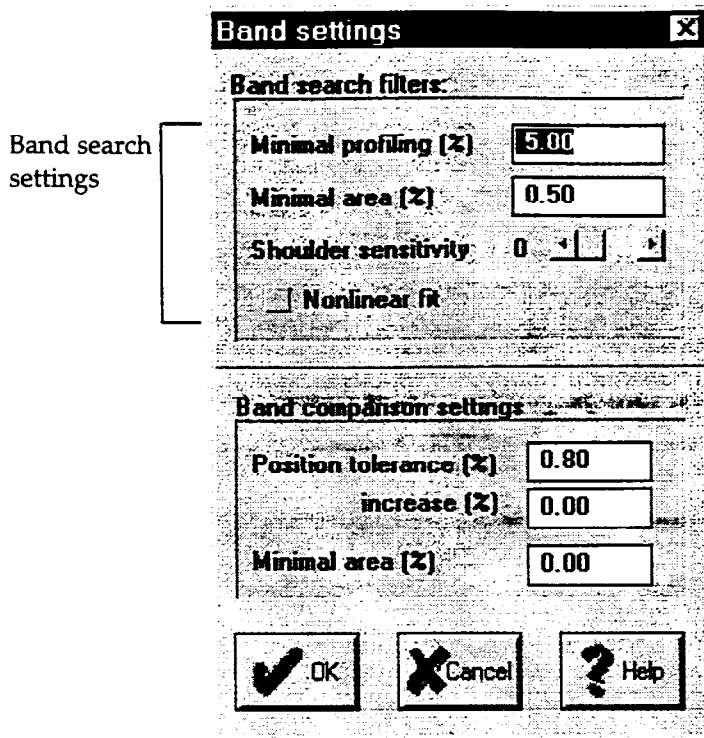


Figure 14. Band settings dialog box. To change search sensitivity, adjust *Minimal profiling* and/or *Minimal area*.

- 7.3.4 Change *Minimal profiling* to adjust the minimum peak height (a percentage of the highest peak of the pattern) and/or *Minimal area* to adjust the minimum area (a percentage of the total area of the pattern). Usually, settings of 5% and 0.5%, respectively, will be convenient. Specify a *Shoulder sensitivity* of 4 to allow the program to find band doublets and bands on shoulders. Click on <OK> to accept the settings.
- 7.3.5 Back in the *Band search window*, you can remove all bands with *Bands\Remove all*.
- 7.3.6 Select *Bands\Auto search* to perform a new band search with the new sensitivity settings.
- 7.3.7 Click on a band to select it, or drag the mouse pointer holding the left mouse button to select a group of bands.
- 7.3.8 Press DEL (or *Bands\Remove*) to delete selected bands.
- 7.3.9 Click the right mouse button to obtain a zoom window of the pointed pattern (Figure 15), where you can add or delete bands at any position using the buttons. Click on <Exit> to exit this window.

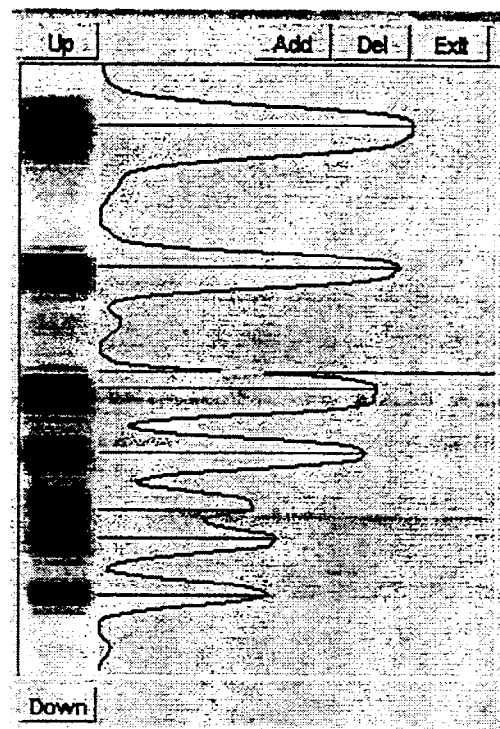


Figure 15. Zoom window in the band search.

7.3.10 Select **<OK>** in the band search window to save the defined bands. The gel window again appears in front.

Repeat steps 7.3.1 to 7.3.10 for other gel files in the database.

7.4 Defining the molecular weight regression

This section outlines the steps for defining a regression for the molecular weight calculation for the whole database. In most cases, this will be a reference pattern.

7.4.1 In the gel window, click a pattern for which you know the molecular sizes of the bands:
Select **TrackDensitometric curve**, which displays a densitometric curve and the gelstrip of the selected pattern (Figure 16).

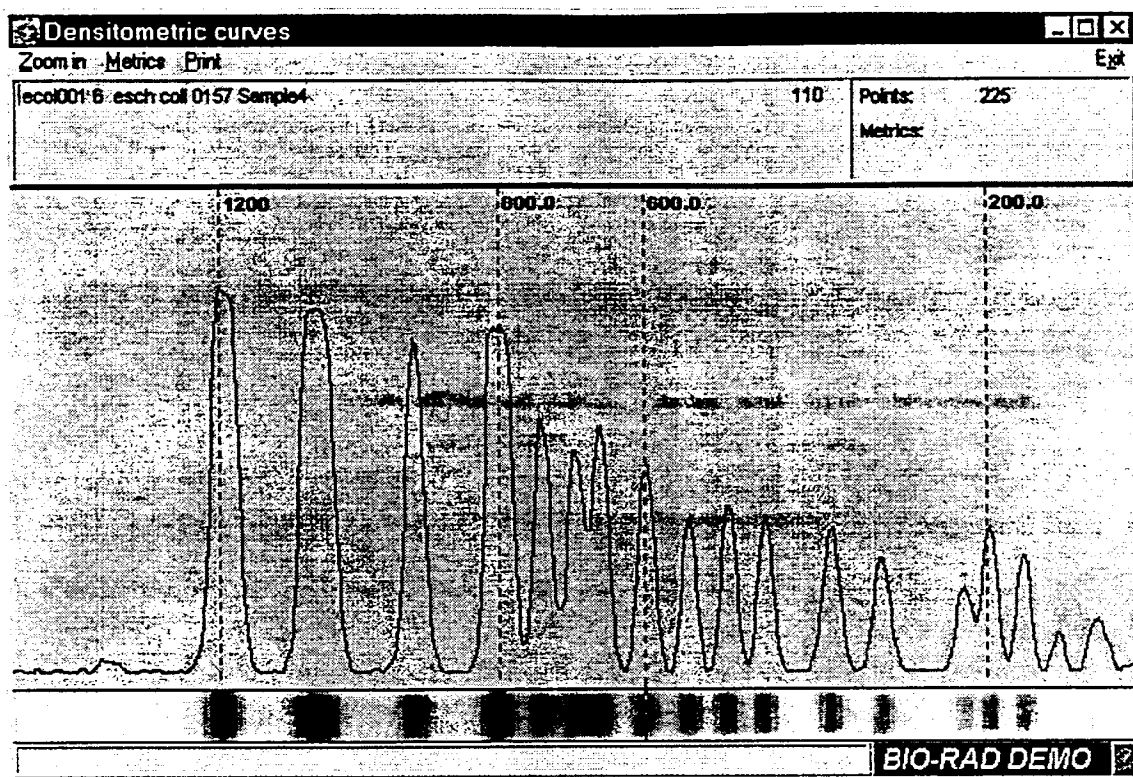


Figure 16. Densitometric curve window.

- 7.4.2 Click on the first known band to position the selector line on it, and specify its size by selecting *Metrics\Add node*.
- 7.4.3 Press <OK> to enter the value and display it on the curve.
- 7.4.4 Repeat steps 7.4.2 to 7.4.3 for all known peaks.
- 7.4.5 Select *Metrics\Interpolate* to define a regression method for the interpolation between the known peaks.

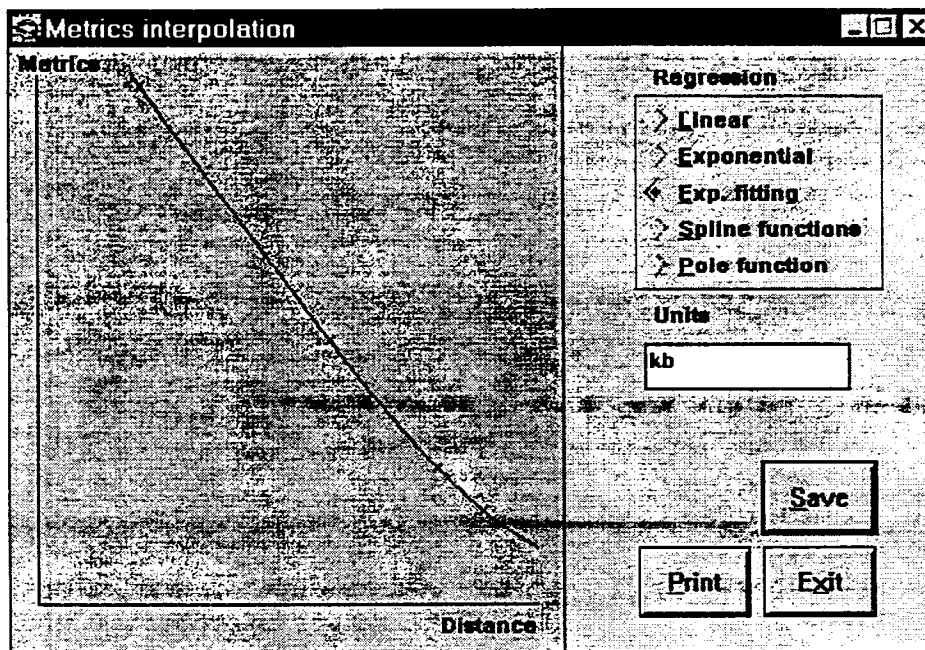


Figure 17. The Metrics interpolation window.

- 7.4.6 Among the different regression methods in the *Metrics interpolation window* (Figure 17), *Exponential* fitting is certainly the most interesting compromise for interpolation and extrapolation (beyond largest and smallest size bands). *Spline functions* offers an exact reproduction of MW on the *nodes* (known bands), but is less reliable for extrapolation.
- 7.4.7 Choose the most suitable regression option and enter the *Units*. (e.g., Kbp, bp).
- 7.4.8 Click on <Save> to save the MW regression for the database. Now all band sizes on all patterns in the database can be quantified.
- 7.4.9 To illustrate this, <Exit> the *Densitometric curve window* and select another lane in the *gel window*.
- 7.4.10 Select *Track/Densitometric curve* again, and click on any band; its size is displayed in the upper right panel of the window.
- 7.4.11 <Exit> this window.
- 7.4.12 Click on <Exit> in the gel window and you are back in the main window.

7.5 Creating lists

Most if not all analysis tools in Fingerprinting Plus are based on selections of patterns from the database, which we call *lists*. The current list of selected patterns is displayed in the right hand panel of the main window (see Figure 11). Initially, there are no patterns, but you can make one as follows:

- 7.5.1 Double-click on a gel file to open it.

- 7.5.2 Click the right mouse button (or SPACE bar) on some of patterns contained in it. They are marked with a purple arrow and will be listed in the list panel if you click on them.
- 7.5.3 Close the gel window by clicking on *<Exit>*.
- 7.5.4 You can also select patterns by a database search. Select *Search\Topic* (or press F5) in the main window. This opens the *Topic search* dialog box (Figure 18).

Figure 18. Topic search dialog box.

- 7.5.5 You can refine the search by filling in a string in one or more of the fields. If you click on *<OK>* without entering a string, you will select all patterns from the database.

Create a list containing all available patterns. We are now ready to start comparisons of the selected patterns.

- 7.5.6 First save the list by selecting *List\Save as*. This opens a *Save list as* dialog box (Figure 19). Enter a name (e.g., "ALL") and click on *<OK>*.

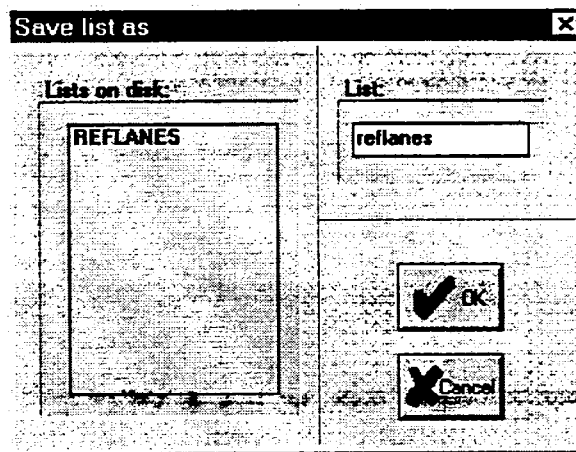


Figure 19. Save list as dialog box.

7.6 Gelstrips and Information boxes

In any window listing gel patterns, you can pop up an *Information box* of a pattern by clicking on the pattern while holding the CTRL key (Figure 20, left). The information box displays all database fields and contains a floating menu, allowing quick-access to the densitometric curve (*Curve*), the gelstrip (*Track*, see below), or the gel file with the lane selected (*File*).

In any window listing gel patterns, clicking on a pattern while holding down the SHIFT key pops up a *Gelstrip* of the selected pattern (Figure 20, right). Multiple gelstrips will automatically align, and they always remain on top of any window. You can drag a gelstrip over the screen to rearrange gelstrips.

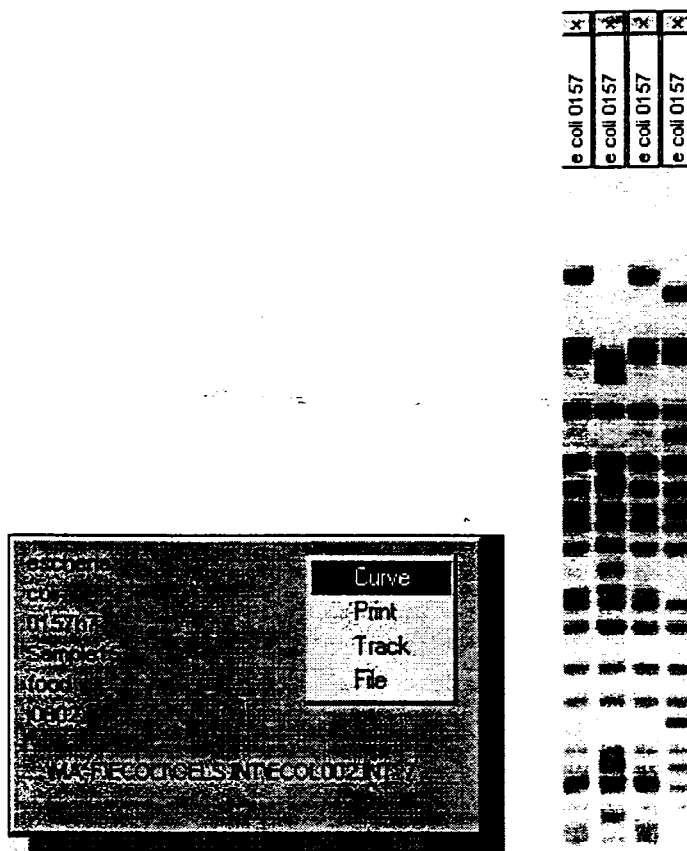


Figure 20. Information box with floating menu (left) and aligned Gelstrips (right).

- 7.6.1 In the list panel (on the right in the main window), click on a pattern while holding down the CTRL key. The *Information box* appears.
- 7.6.2 Select *Curve* from the floating menu. This opens the densitometric curve window.
- 7.6.3 *Exit* the densitometric curve window.
- 7.6.4 In the list panel again, click on several patterns while holding down the SHIFT key. Their *Gelstrips* appear aligned on the screen.
- 7.6.5 Select *Gelstrips\Labels\Entry code* in the main menu. The gelstrips are now marked with the entry codes.
- 7.6.6 Drag some of the gelstrips over the screen by holding down the left mouse button as you move the mouse.
- 7.6.7 Select *Gelstrips\Rearrange* in the main menu to realign the gelstrips in the new order.
- 7.6.8 Click the close button ☐ in the upper right corner to remove a gelstrip.
- 7.6.9 Select *Gelstrips\Remove all* to remove the remaining gelstrips.

Both information boxes and gelstrips can be called in virtually every window listing patterns: gel windows, lists, dendrograms, principal component analyses, all identification windows, and comparative quantifications. Be sure to use these wonderful tools!

7.7 Composed images

You can display composed gel images of any selection (list) of database patterns (Figure 21).

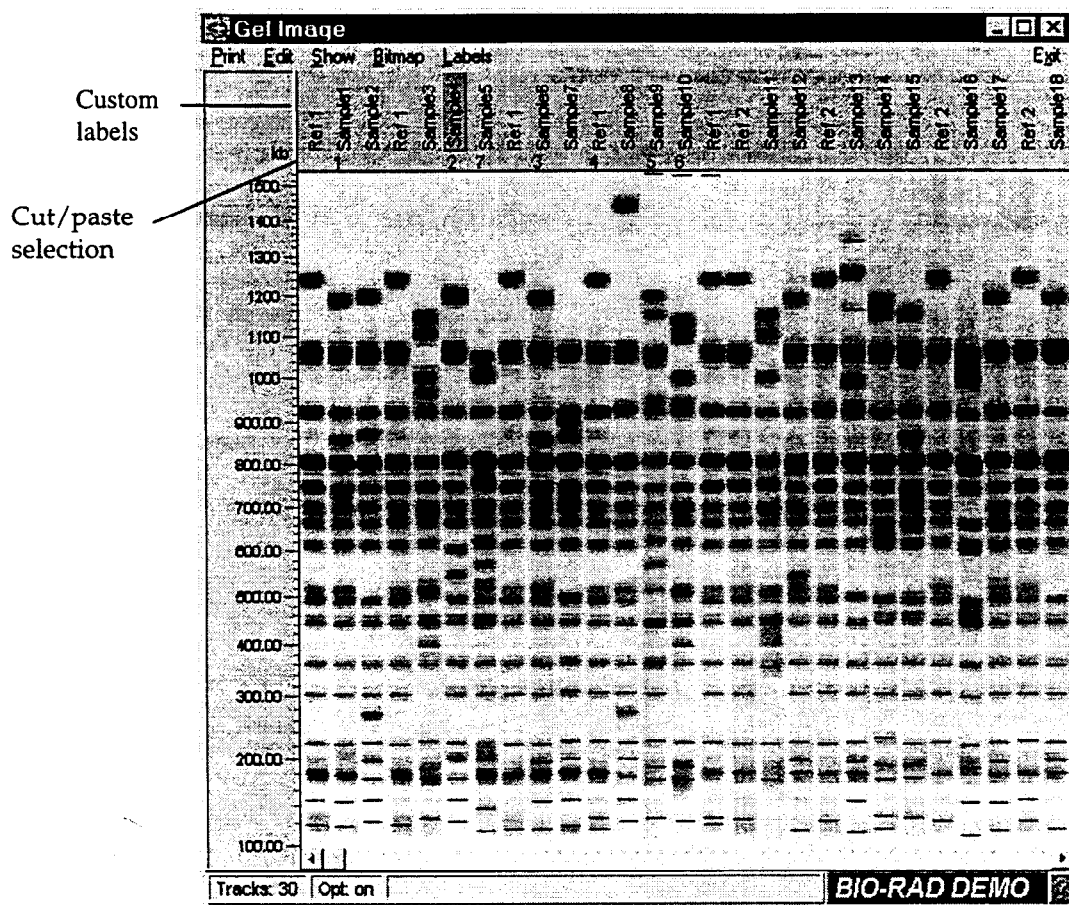


Figure 21. Gel image of selection of database patterns.

- 7.7.1 Select *Edit2D-Gel image* in the main window. The image shows normalized gelstrips of the selected patterns.
- 7.7.2 Select *ShowBands* and *ShowMetric scale* to view the bands defined for the patterns and the molecular weight scale.
- 7.7.3 Add custom labels to the gelstrips with the *Labels* command (see 7.8.7 to 7.8.10).
- 7.7.4 Click the *right* mouse button on some gelstrips to select them sequentially (1, 2, 3 etc. on Figure 21).

7.7.5 Select **Edit|Cut** to cut the selected lanes from the image.

7.7.6 Now click any pattern (with the left mouse button) and select **Edit|Paste**. The patterns cut from the image are pasted *after* the selected pattern, in the order of selection.

You can print the composed image by selecting **Print**. You can create a bitmap file by selecting **Bitmap**. Fingerprinting Plus automatically opens Paintbrush with the created bitmap

7.7.7 Close the composed image by clicking on **<Exit>**.

7.8 Cluster Analysis

Internally, the computer calculates a dendrogram in two steps. First it calculates a *similarity matrix* between all possible pairs of patterns from the selection list by using a certain *similarity coefficient*. Then it uses this matrix to calculate a *dendrogram* by a *clustering technique*.

One method of calculating similarity between patterns is by comparing the similarity between their densitometric curves using a correlation coefficient. To use this method, select **Comparison|Clustering (correlation)** from the main menu. Another method is by comparing the bands present in two patterns and using the fraction they have in common as the basis for similarity. This method only makes sense if you have defined the bands for all the patterns of the selected list. To use this method, select **Comparison|Clustering (bands)**.

7.8.1 For this example, begin a cluster analysis by selecting **Comparison|Clustering (correlation)**. This opens the *Grouping analysis* dialog box (Figure 22), prompting for a *clustering method*. Refer to the manual for an explanation of the different clustering methods.

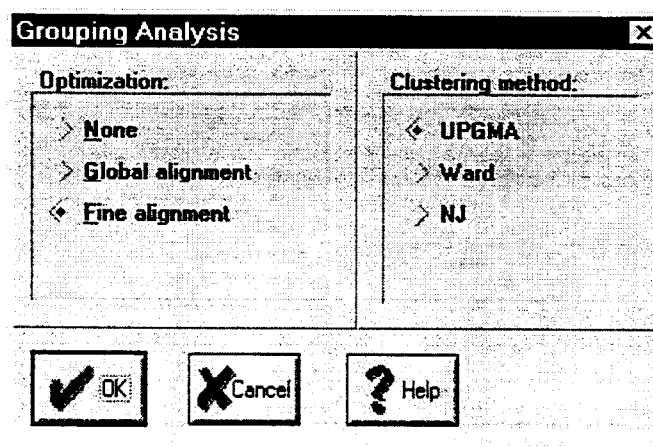


Figure 22. Grouping analysis dialog box.

7.8.2 **UPGMA** is the most common method; select this one.

7.8.3 Under *Optimization*, select **Fine alignment** to obtain an ultimate trace-to-trace optimization during the calculation of the coefficient.

7.8.4 Click on **<OK>** to begin the cluster analysis. The result is a *Dendrogram window* (Figure 23).

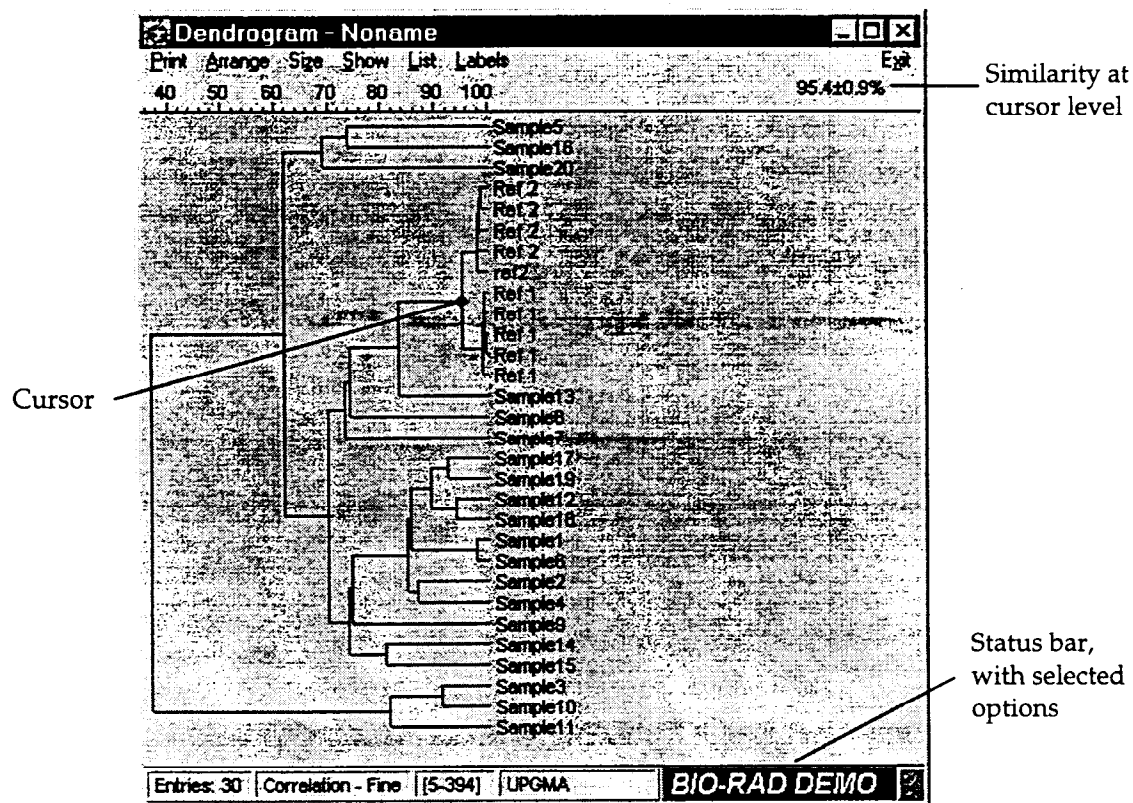


Figure 23. Dendrogram window.

7.8.5 You can move the cursor to any branch using the arrow keys or by clicking on the branch.

7.8.6 To show the image of the patterns next to the dendrogram, select *Show\Gelstrips*.

By default, the gel name and lane number are displayed for each pattern. You can also custom-define your own combination of information fields.

7.8.7 Select the *Labels* menu (press F9 from any window, or select *System\Names* from the main window). This opens the *Entry description* dialog box (Figure 24).

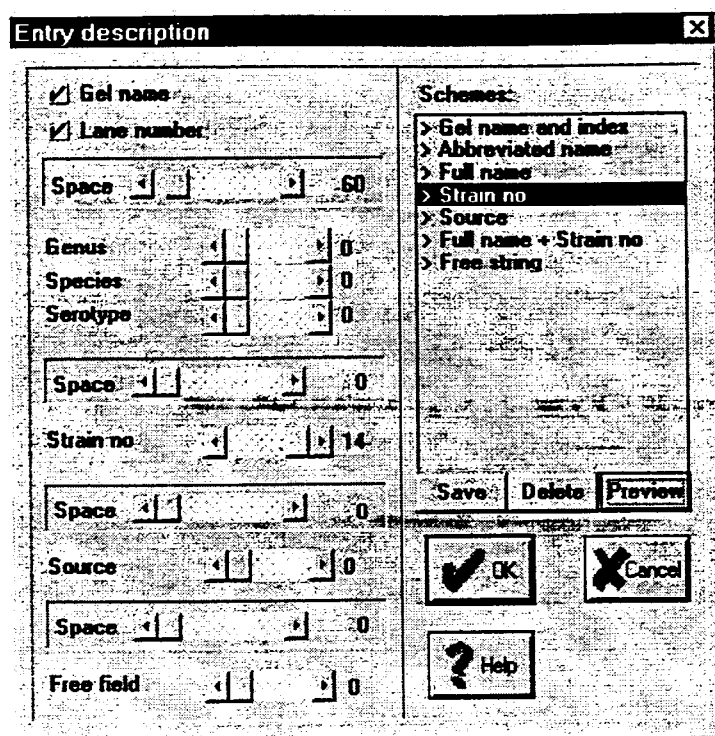


Figure 24. Entry description dialog box.

- 7.8.8 Choose one of the predefined combinations and click on **<Preview>** (move the dialog box away from the center if necessary).
- 7.8.9 You can add or delete information fields up to a fixed number of characters and increase or decrease space as needed.
- 7.8.10 Click on **<Save>** to save your combination or **<OK>** to implement your combination.
- 7.8.11 Back in the Dendrogram window, display the defined bands on the patterns with **ShowBands** and show a molecular weight scale on the patterns with **ShowMetrics**.

There are two ways to print a dendrogram: a standard mode (**PrintFull size** and **PrintFit on page**), which creates a print job in the printer manager; and a WYSIWYG mode (**PrintPreview mode**), which works with a page preview editor and prints a dendrogram directly to the default printer device. The latter method is the recommended mode for publication quality.

- 7.8.12 Select **PrintPreview mode** to see the dendrogram in the *Preview mode window* (Figure 25).

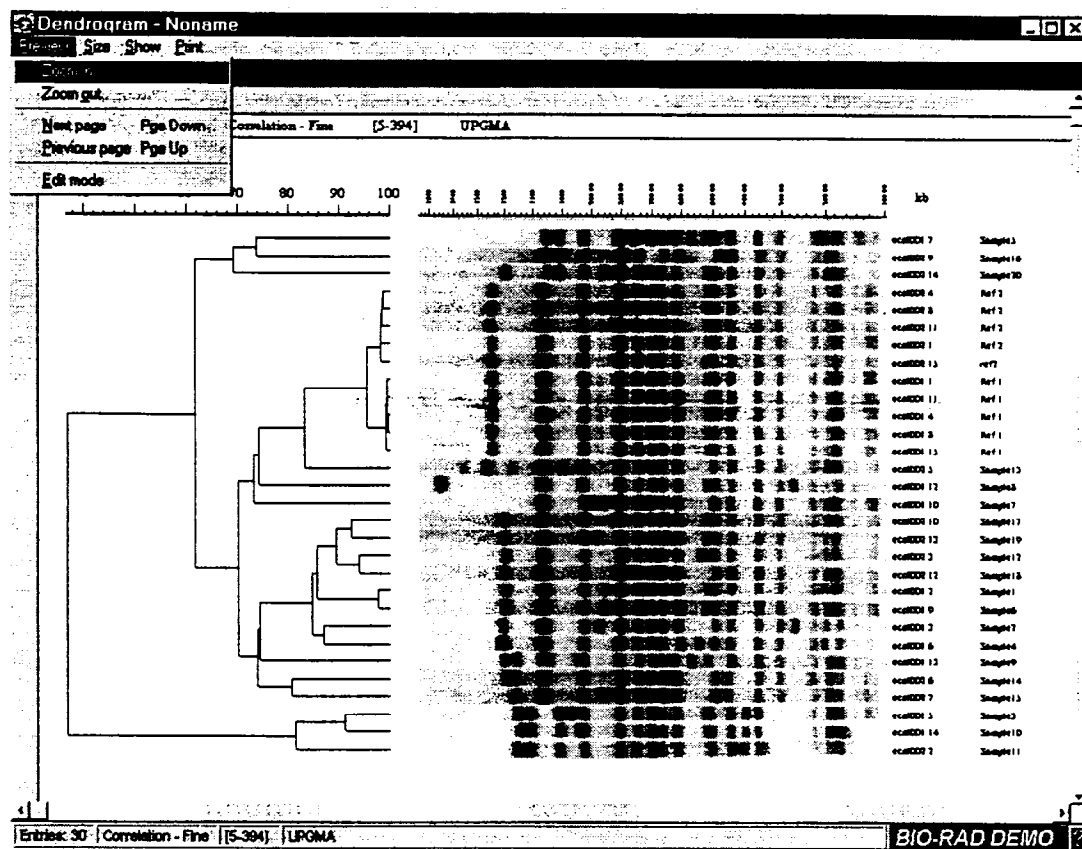


Figure 25. Dendrogram window in print preview mode.

- 7.8.13 In preview mode, you can **Zoom in** or **Zoom out**, move to **Next page** and **Previous page**, adjust the image **Size** between **1.0 mm** and **8.0 mm**, or have the dendrogram **Fit on page**.
- 7.8.14 **Print**/**Print all pages** prints the dendrogram.
- 7.8.15 Select **Preview**/**Edit mode** to return to the edit mode of the dendrogram window.
- 7.8.16 Back in edit mode, **Show**/**Matrix** displays the similarity matrix of the dendrogram. The **Matrix window** (Figure 26) represents the similarity values as differentially shaded blocks.
- 7.8.17 You can see the original similarity values by selecting **Layout**/**Show numerical values**.

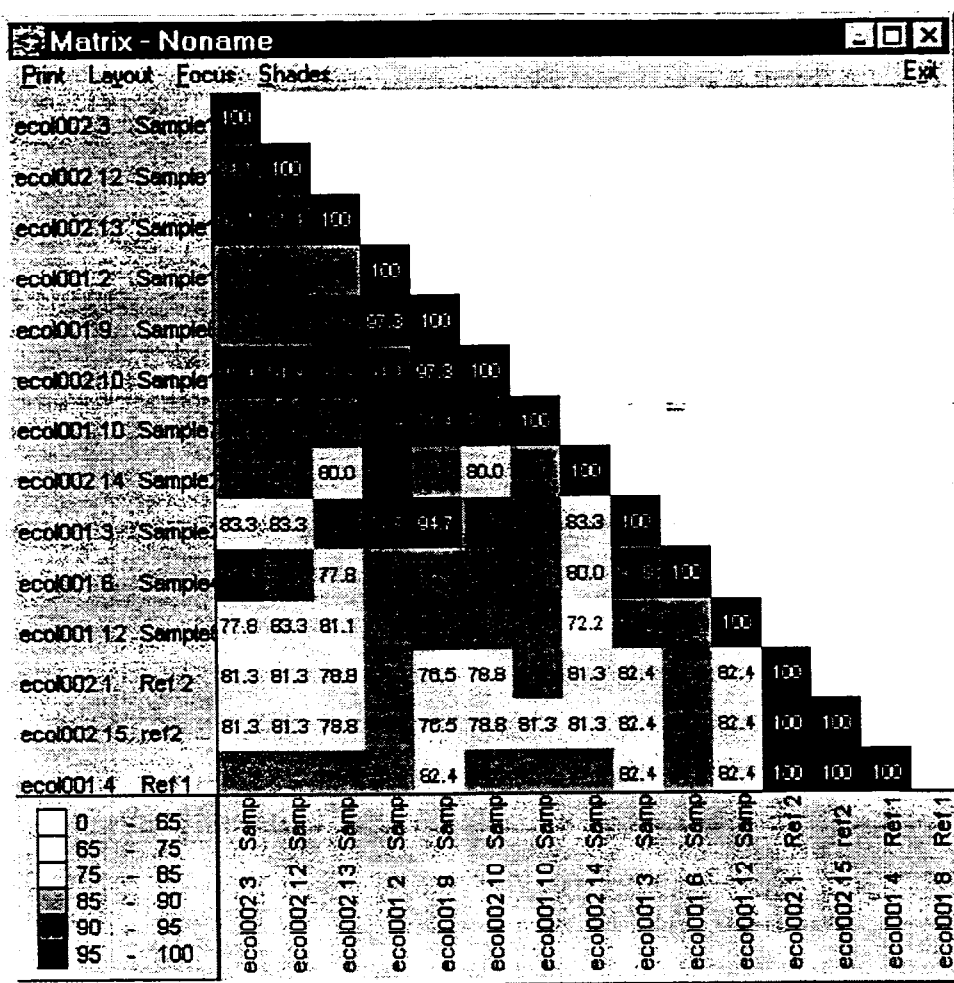


Figure 26. Similarity matrix of a cluster showing numerical similarity values.

7.8.18 The **Print** menu allows you to print *Full size*, *Fit on page*, or *Print numerical values*.

7.8.19 A file named COR.TXT is created with **Print|Export numerical values**.

7.8.20 Closing the *Matrix window* and the *Dendrogram window* puts you back in the *Main window*.

You can cluster the same patterns using a *band matching coefficient*.

7.8.21 Select **Comparison|Clustering (bands)**. This opens another grouping analysis dialog box, for band-based coefficients (see Figure 27).

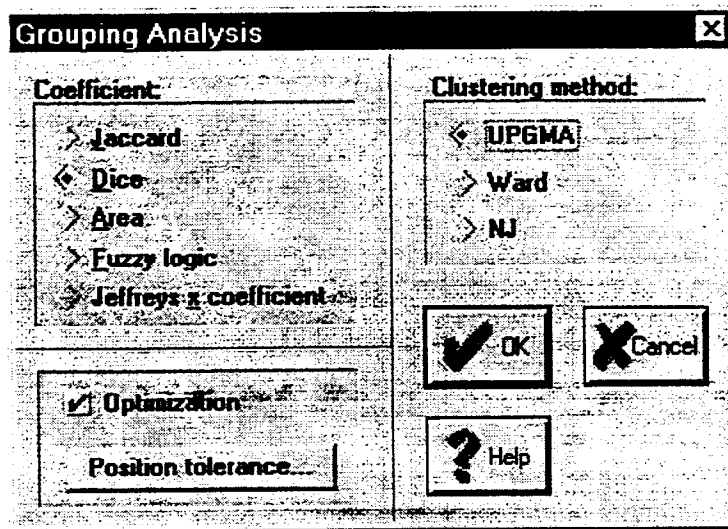


Figure 27. Grouping analysis dialog box for band-based similarity coefficients.

7.8.22 The *Dice* coefficient is the most common; select this option. See the reference manual for more information on the different coefficients.

7.8.23 Select *UPGMA* clustering.

7.8.24 Enable *Optimization*. Trace-to-trace optimization compensates for remaining shifts between patterns.

An important adjustable parameter in band-based comparisons is the *position tolerance*—the tolerance allowed for two bands with slightly different positions to still be considered matching. It is absolutely necessary to allow a certain tolerance, but the amount depends on the quality and nature of the patterns.

7.8.25 Click on <Position tolerance> to display the *Band settings* dialog box (see also 7.3.3).

7.8.26 In the lower panel, *Band comparison settings* (Figure 28), select a *Position tolerance* of 1.00 (percent of total track length), a *Position increase* of 0.0 (toward the bottom of the patterns), a *Minimal peak area* of 0.0 (percent of the total pattern area), and a maximal *Optimization* shift of 0.50% (percent of the pattern length).

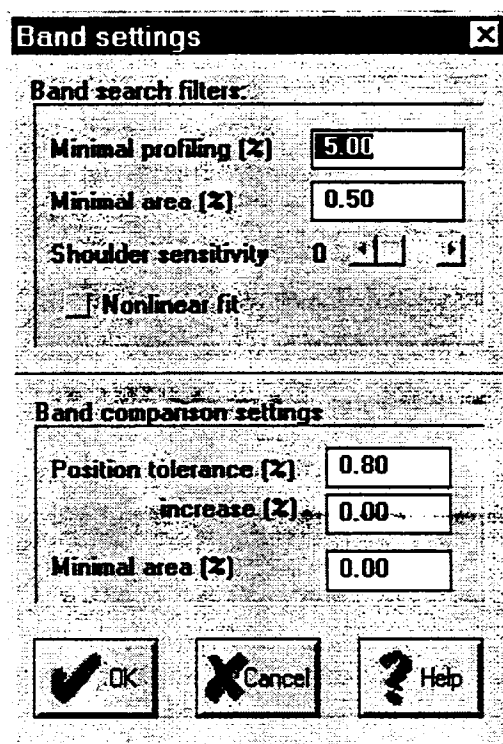


Figure 28. Band settings dialog box: adjustable band comparison settings in lower panel.

7.8.27 Click on <OK> to accept these settings.

7.8.28 Click on <OK> in the *Grouping analysis* dialog box to begin the clustering. Further functions are the same as those described in 7.8.5–7.8.20.

7.9 Identification

Fingerprinting Plus offers three identification concepts:

- 1) *List identification*. A gel lane or list of lanes is identified based on another, saved list. The latter list may include the whole database or a subset thereof.
- 2) *Library identification*. One or more *libraries* can be created, each constituting representative patterns for a biological entity. Unknown patterns are identified against a library.
- 3) *Identification groups*. These are specific groups of patterns representing a biological entity, for which the frequencies of all bands are calculated. Unknown patterns are identified in a more statistical way, taking into account the relevance of each band based upon its frequency.

In this guide we will only consider the *list* method. For example, if the database contains bacteria belonging to the genera *Pseudomonas* and *Stenotrophomonas*, you can create separate lists for these genera using the search function (7.5.4) and save these to disk (7.5.6). A third list may contain all database patterns (except the reference patterns, if these are size markers). Then, if you already know to which genus an unknown strain belongs, you can identify it against the list containing that one genus. Or if you know nothing about the strain, you can identify it against the list ALL.

- 7.9.1 Make sure that no list is selected. If the right panel shows selected patterns, choose **List|Clear list** in the main window.
- 7.9.2 Double-click on a gel to open it.
- 7.9.3 Select a few lanes on the gel (click with the *right* mouse button).
- 7.9.4 Close the gel ☐.
- 7.9.5 Select **Comparison|Identification with list**. The *Identification with list* dialog box opens, showing all available saved lists and prompting for a similarity coefficient (Figure 29).

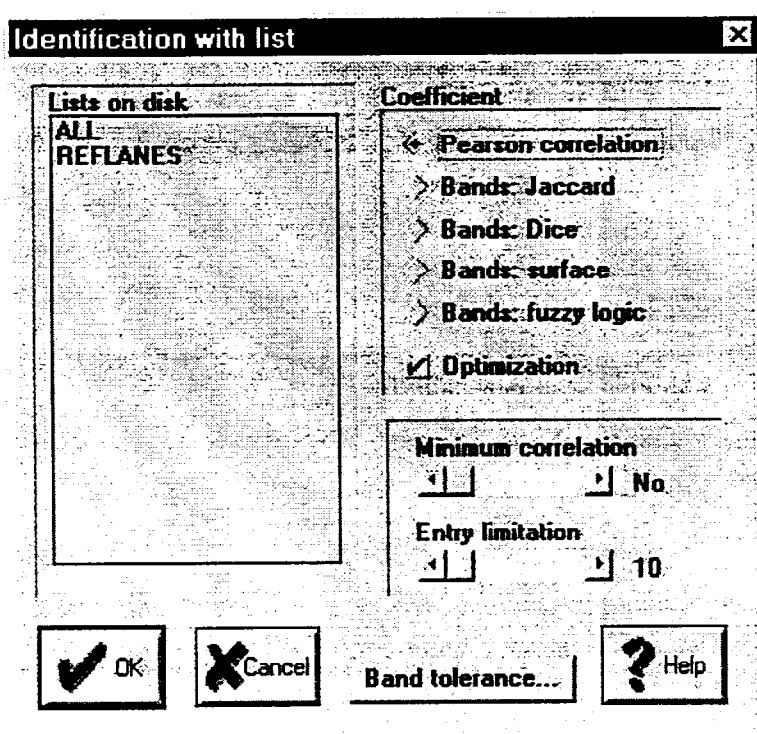


Figure 29. Identification with list dialog box.

- 7.9.6 On the left are the lists against which the unknown pattern(s) will be identified. Select one of them (e.g., ALL).
- 7.9.7 Select a coefficient: The *Pearson correlation* is curve-based; the others are band-based. For this example, select *Pearson*.
- 7.9.8 Enable the *Optimization*.
- 7.9.9 *Minimum correlation* specifies that only identification cases above a certain similarity will be shown. Leave this parameter at zero (no minimum).
- 7.9.10 *Entry limitation* specifies that only the 10 best matching database patterns will be displayed for each unknown pattern. Up to 100 best matching cases can be displayed, but we keep the limitation of 10 here.

7.9.11 Click on **<OK>** to start the identification.

Once identification is complete, the *Identification report window* (Figure 30) opens. The unknown patterns are listed on the left, and the best matching database entries are shown on the right. In this example, since the unknowns themselves are present in the list ALL, they are listed as best matching with 100% similarity.

7.9.12 Press the F9 key to call the *Entry description* dialog box. Select (for example) **Full name + Label 1** and click on **<OK>**. This changes the description of the best matching patterns as shown in Figure 30.

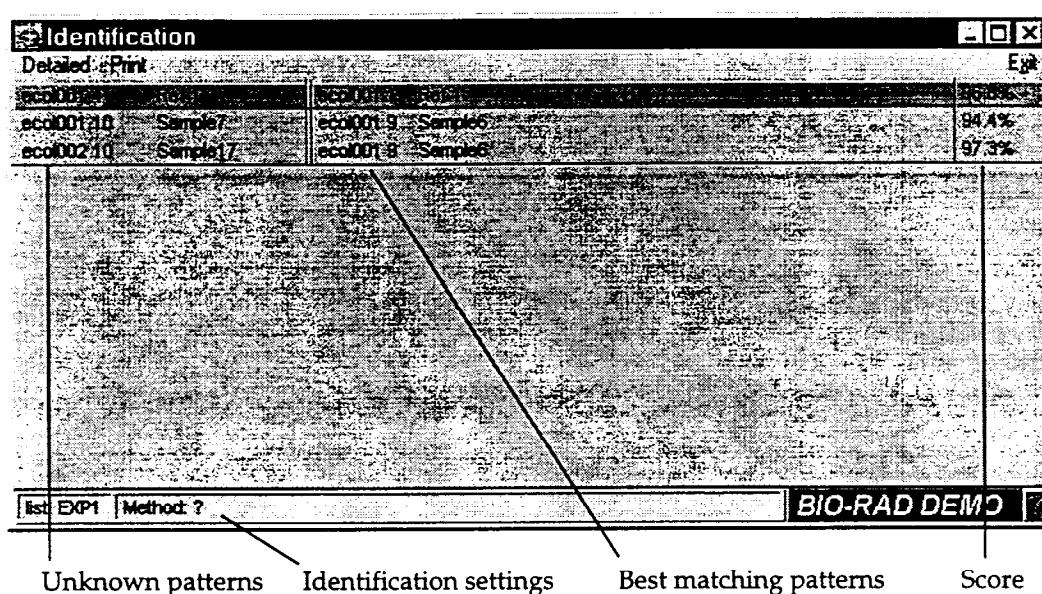


Figure 30. Identification report window.

7.9.13 **Print** the identification report.

7.9.14 Click on one of the unknowns and select **Detailed** to see the 10 best matching database patterns with the selected unknown (Figure 31).

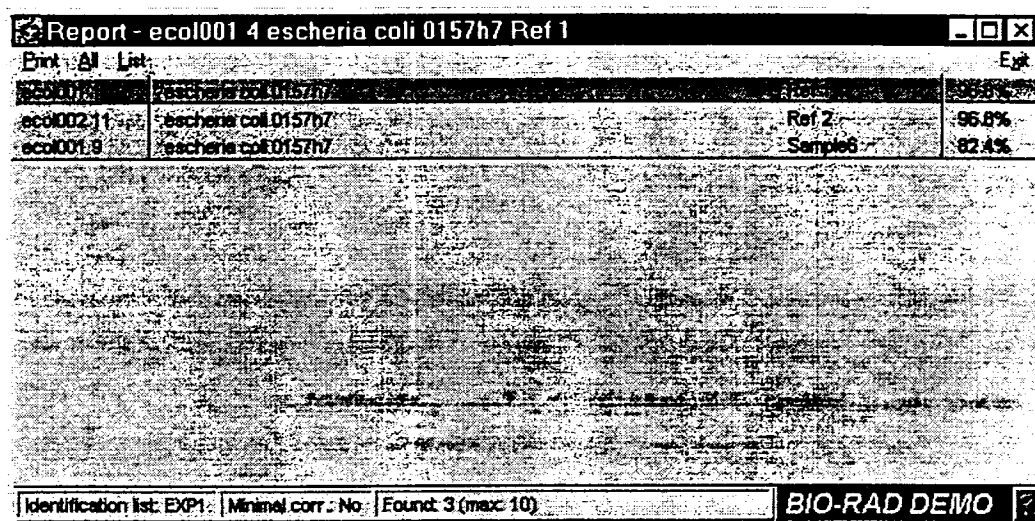


Figure 31. Detailed identification report.

The database patterns are shown in decreasing order of similarity. You can also display the gelstrips.

7.9.15 In the *Identification report window*, hold down the SHIFT key while clicking on the selected unknown. This displays the gelstrip of the *unknown* on the screen.

7.9.16 In the *Detailed identification report*, hold down the SHIFT key while clicking on the best matching database patterns. Their gelstrips are shown next to the unknown gelstrip (Figure 32).

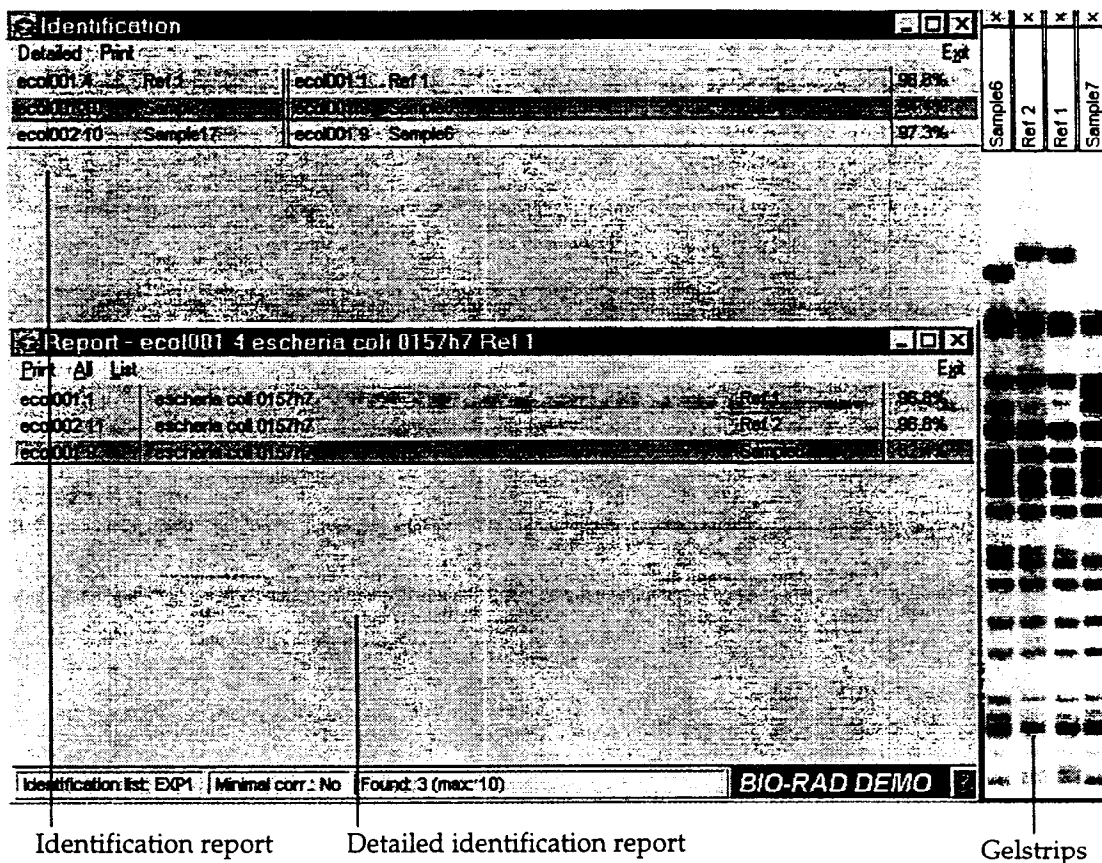


Figure 32. Identification report and detailed identification report. Gelstrips for the unknown are on the right.

7.9.17 Click the close window button  for each gelstrip.

7.9.18 **Print** the detailed identification report.

7.9.19 **Exit** the detailed identification and **Exit** the identification window.

8. Printing

Most functions in Fingerprinting Plus create *print jobs* in the *Printer manager* rather than sending the output directly to the printer. This has several advantages: (1) Print jobs can be saved to a disk and exchanged between different sites; (2) print jobs remain available until you delete them, can be printed all at once, and can be printed on different printers, with different printer settings etc.; and (3) the printer manager can export all print jobs as Windows clipboard files.

8.1.1 In the main window, select **System|Printer manager** (or press the F8 key from within any subwindow). This calls the *Printer manager* (Figure 33, right). The job names and the date and hour of creation are shown.

8.1.2 Click on a print job (e.g., Tree) and select **Preview**. A page preview of the selected job is shown (Figure 33, left).

8.1.3 Select *PgeDown* and *PgeUp* if the job requires more than one page.

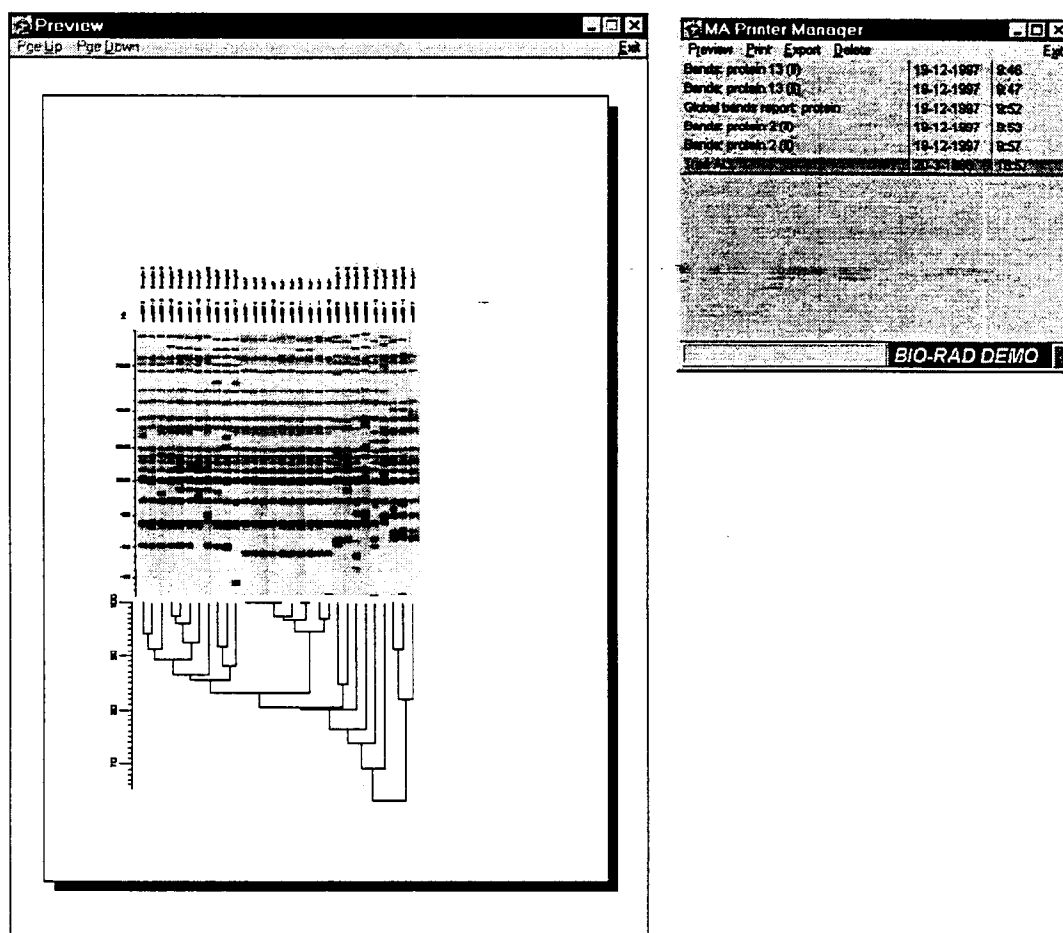


Figure 33. Printer manager with preview of selected job.

8.1.4 With *Export|To clipboard*, you can *Edit|Paste* the image in any graphical Windows program.

8.1.5 *Exit* the preview and select *Print|Print selected job*.

8.1.6 When printing is finished (i.e., when the progress indicator disappears), you can delete the print job with *Delete*.

8.1.7 *Exit* the *Printer manager*.